

Interactive graphics: Pointing and positing devices (cursor, lightpen, digitizing tablet, the mouse, track balls). Interactive graphical techniques. Positioning (Elastic or Rubber Bank lines, Linking, zooming, panning clipping, windowing, scissoring). Mouse Programming.

Interactive Graphics

Interactive graphics is a branch of computer graphics that allows users to **communicate with computers through graphical interfaces**. Unlike static graphics, interactive graphics respond to **user inputs** provided via devices like the mouse, light pen, or tablet. Such interactivity enables drawing, editing, modeling, and visualization in real time.

1. Pointing and Positioning Devices

Pointing devices allow users to interact with graphical objects by selecting, drawing, and moving them within the display screen.

a. Cursor

- The cursor is a **symbol displayed on the screen** to indicate the current position where user input will be applied.
 - It can take forms like an arrow, crosshair, or blinking line.
 - Controlled by devices such as the mouse, trackball, or keyboard.
 - Essential in positioning tasks like selecting menu items, editing text, or drawing graphics.
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b. Light Pen

- A light-sensitive pointing device shaped like a pen.
 - When the tip of the pen touches the screen, it detects the electron beam of CRT (Cathode Ray Tube) displays.
 - The computer calculates the pen position by sensing the time at which the beam passes.
 - Used for **object selection and drawing directly on the screen**.
 - Limitations: Only works effectively with CRT monitors, not modern flat panels.
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c. Digitizing Tablet

- Also called a **graphics tablet** or **digitizer**.
 - Consists of a flat surface and a stylus (pen-like device).
 - User draws on the tablet, and the position is transmitted to the computer.
 - Used in CAD (Computer-Aided Design), GIS (Geographical Information Systems), and animation.
 - Supports pressure sensitivity, enabling variable line thickness.
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d. Mouse

- A small handheld device with buttons and a scroll wheel.

- Movement of the mouse corresponds to the movement of a pointer on the screen.
 - Types: Mechanical (with a ball), Optical, and Laser.
 - Common functions: Clicking, dragging, double-clicking, and scrolling.
 - Widely used in GUIs for selecting objects, menu navigation, and drawing.
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e. Trackball

- A pointing device consisting of a **ball fixed in a socket**.
 - The user rotates the ball with fingers or palm, and sensors detect movement.
 - Functions similarly to a mouse but stationary (ball moves instead of the device).
 - Often used in laptops, aircraft cockpits, and specialized CAD terminals.
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2. Interactive Graphical Techniques

Interactive graphics techniques enable users to **manipulate and control objects** in real-time using pointing devices.

a. Positioning

- The process of specifying the **location of graphical objects**.
 - Techniques include:
 - **Elastic or Rubber Band Lines**
 - Used for drawing straight lines or shapes interactively.
 - The line stretches like a rubber band from a fixed starting point to the current cursor position.
 - Common in drawing software and CAD systems.
 - **Linking**
 - Establishing relationships between objects.
 - Example: Linking text to a graphical shape or connecting blocks in a flowchart.
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b. Zooming

- Technique to **enlarge or shrink** a portion of the display.
 - Helps users focus on details (zoom in) or view the entire structure (zoom out).
 - Example: Zooming into a map in GIS software or into an image in Photoshop.
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c. Panning

- Technique to **shift the view of a display window** horizontally or vertically without changing zoom level.
 - Similar to moving a camera across a scene.
 - Example: Scrolling across a large drawing in AutoCAD.
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d. Clipping

- Process of **removing portions of a picture** outside a specified region (called a clipping window).
 - Ensures that only visible parts of objects are displayed.
 - Clipping can be applied to points, lines, polygons, or text.
 - Example: Line clipping algorithms (Cohen-Sutherland, Liang-Barsky).
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e. Windowing

- Refers to **selecting a portion of a graphical scene** (a window) for display.
 - Acts like a virtual camera focusing on part of the scene.
 - Different from zooming: windowing restricts the visible region, while zooming changes scale.
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f. Scissoring

- Special type of clipping where a rectangular portion of the display is defined as a "scissor box."
 - All drawing outside the scissor box is ignored.
 - Used in interactive applications for defining working areas.
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3. Mouse Programming

Mouse programming refers to **handling mouse input in interactive graphics applications**.

Functions of Mouse in Programming

- **Movement Detection**
 - Captures cursor coordinates (x, y) when the mouse is moved.
- **Click Detection**
 - Single-click, double-click, right/left button click are detected.
- **Drag and Drop**
 - Mouse movement while holding a button is used for selecting and moving objects.
- **Scrolling**
 - The scroll wheel generates events to zoom or scroll in applications.

Mouse Programming in Graphics Applications

- Programming languages such as **C, C++, Java, Python** provide mouse event handling functions.
- Event-driven programming captures mouse events (e.g., `onClick()`, `onMouseMove()`).
- Example in C/C++ (with graphics libraries):
 - `getmousepos(&x, &y);` – Gets current mouse position.
 - `mouseclick()` – Detects button press.

Applications of Mouse Programming

- Graphical User Interfaces (Windows, MacOS, Linux).
- Drawing software (AutoCAD, Photoshop, MS Paint).
- Video games (character control, aiming).
- Simulation environments (3D modeling).